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PEARL

Differentially Private and Entropy-Aware Regulated Language Generation

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The Blind Spot of DP: Trustworthiness

RAG improves factuality but widens the privacy attack surface — DP is the standard safeguard

What prior DP work misses

Focus only on privacy–utility trade-off

Report accuracy or perplexity at a given privacy budget; trustworthiness is left underexplored

Hallucination — a core threat —

gets little explicit attention under DP

What we show

DP noise amplifies knowledge conflict

and thus increases hallucination

Yet privacy can be preserved while

hallucination is reduced

(demonstrated by PEARL)

Key Accounting for hallucination when applying DP removes inefficiencies of the usual utility trade-off.

Why DP Worsens Hallucination

Proposition 1 — knowledge conflict amplification under noise

Setup

- p : token probs with retrieval
- q^{para} : token probs query-only (no retrieval)
- $\tilde{p} \propto exp(\beta \cdot \phi)$: exponential-mechanism noise
  $0 < \beta < 1$ shrinks as privacy tightens
- Smaller β flattens p → raises entropy $H(\tilde{p})$
- while $H(q^{para})$ stays fixed

The entropy gap is amplified

$$\log\left[\frac{\mathrm{H}(\mathrm{q}^{\mathrm{para}}) - \mathrm{H}(\tilde{\mathrm{p}})}{\mathrm{H}(\mathrm{q}^{\mathrm{para}}) - \mathrm{H}(\mathrm{p})}\right] > 0$$

given $H(q^{para}) - H(p) < 0$

DP noise widens the gap vs. the parametric prior

Takeaway As the gap widens, decoding drifts toward prior-driven, unsupported tokens — i.e., more hallucination.

Key Signal: the Confidence Gap (CG)

$CG = H(p) - H(\tilde{p})$: entropy difference with vs. without retrieved context

Privacy-related content

PII / quasi-identifiers tied to a person

PII-bearing tokens show HIGH CG

entropy drops sharply when the model regurgitates retrieved context

→ abstain / mask

General content

Impersonal factual content

Hallucinated tokens show LOW CG

weak grounding in retrieved context raises risk of non-faithful output

→ stabilize / regenerate

Key CG is a direction-sensitive signal: low CG flags hallucination, high CG flags PII leakage.

CG Actually Separates the Two Risks

LeakRAG benchmark · LLaMA 3.1 8B responses labeled per sentence by GPT-4o + domain experts

① Hallucination

Supported sentences
→ clearly higher CG distribution

Unsupported (hallucinated)
→ lower CG distribution

more knowledge conflict ⇒ more hallucination

② PII Leakage

PII-bearing sentences
→ higher top-5 token CG

non-PII sentences
→ relatively lower CG

uncertainty drops when regurgitating context

Insight Hallucination sits at low CG, PII at high CG — opposite directions, so one signal catches both.

PEARL: a Two-Step, CG-guided Framework

Entropy-aware private decoding — spend the budget where it matters most

Step 1 · Initial response

① DP decode

exponential mechanism sampling

Step 2 · CG-guided regeneration

② Compute CG

context $\tilde{p}$ vs base p entropy difference

③ Filter

Top-k selects PII Sp and hallu. Su

④ Refill

redact Sp, refill Su via keywords

Key idea

Rather than spreading noise uniformly, PEARL concentrates the privacy budget on critical spans.

Privacy budget allocation

Generation $\leq$ 80% · filtering + refill split evenly — composed in RDP via autoDP, then converted to (ε, δ)-DP

The LeakRAG Benchmark

First privacy-focused RAG evaluation in medical and financial domains, with explicit PII annotations

1. **Explicit PII annotations**
   entity-level labels enable controlled leakage measurement

2. **Synthetic PII**
   real identifiers infeasible under HIPAA / GDPR → expert-validated synthetic PII

3. **Expert-verified documents**
   domain experts (e.g. a bank branch manager with 32 years of service) confirm realism

Medical
1,200 docs · avg 10.4 PII / doc (ChatDoctor)

Financial
286 docs · avg 3.3 PII / doc (Banking77)

Main Results

$\varepsilon \in \{3, 6, 8\}$ — LLaMA 3.1 8B · LeakRAG Medical (vs. NoRefill / RandomRefill)

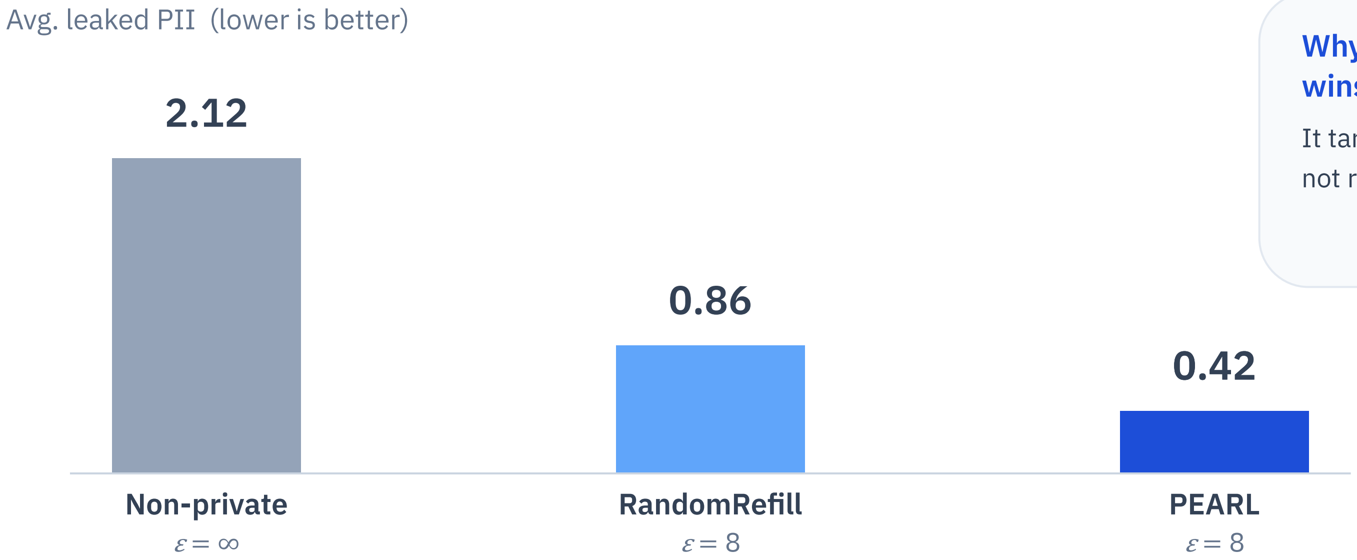
Method	BERTScore ↑	GoldAlign ↑	HalluScore ↓
NoRefill	50.6 / 52.9 / 52.9	2.53 / 2.80 / 2.88	2.52 / 2.19 / 2.15
RandomRefill	57.0 / 58.6 / 58.6	3.01 / 3.09 / 3.06	1.97 / 1.92 / 1.90
PEARL	**58.5 / 58.5 / 58.6**	**3.06 / 3.08 / 3.14**	**1.87 / 1.90 / 1.87**

Values shown in $\varepsilon = 3$ / 6 / 8 order

Key PEARL lowers HalluScore (up to 0.22) and raises GoldAlign while keeping BERTScore near the non-private level.

Defense Against Privacy Attacks

Malicious PII-extraction prompt — avg. number of leaked PII (LeakRAG Medical, LLaMA 3.1 8B)



Why PEARL wins

It targets sensitive spans via CG, not random selection

Result CG-based redaction of sensitive spans yields consistently lower PII leakage than random selection.

Conclusion

Entropy-aware DP decoding that addresses privacy and hallucination together

1. **DP worsens hallucination**
   noise widens knowledge conflict and amplifies hallucination (shown in Section 3)

2. **CG links the two risks**
   the confidence gap correlates with hallucination and PII leakage — with direction

3. **PEARL: CG-guided rewriting**
   regenerate hallucinated spans, redact PII spans — budget focused where it matters

4. **Privacy kept, hallucination cut**
   stronger robustness to PII-extraction attacks; hallucination down, utility preserved

→ Supporting safer LLM deployment in privacy-sensitive domains such as healthcare and finance